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FEATURE ARTICLE

INTERNATIONAL

The Effects of Web-Based Diabetes Education on Diabetes Care Results

A Randomized Control Study

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This study aimed to test the effects of providing Web-based diabetes education to individuals with type 2 diabetes on the A_{1c} level and health check attendance. The study participants comprised 122 individuals with type 2 diabetes, who had access to the Internet, had completed their basic diabetes education, and had similar basic situational factors. Using a randomization method, these participants were chosen from the patients being monitored by the diabetes nurses. The experimental group (n = 61) was monitored via the Web. From measurements recorded in the sixth month of monitoring, we found that A_{1c} levels of the individuals with diabetes who were monitored through the Web decreased ($t = 6.63$; $P < .05$), and the rate of attending health check visits increased ($z = 5.97$; $P < .05$), while no difference was detected in the control group ($t = -0.63$; $P = .534$; $z = -0.80$; $P = .426$). To maintain glycemic control, Web use could be adopted as a complementary tool for monitoring individuals with diabetes.

KEY WORDS

A_{1c} • Diabetes education • Health check • Internet • Web-based education

Diabetes is a chronic disease that results in morbidity and death. It is predicted that the incidence of diabetes, which is one of the most common chronic diseases, will increase rapidly worldwide in the next 20 years, and 171 million people will have diabetes.[1] According to the results of the Turkey Diabetes Epidemiology Study reported by Satman et al,[2] in Turkey, there are 2.6 million people with diabetes and 2.4 million people with glucose intolerance. In diabetes management, education of the individual with diabetes and continuity of this education are important.[3,4] In recent years, Internet use has become a more common avenue for the education of individuals with diabetes because information, when needed, is easily accessible and the cost is low; thus, Internet use has been encouraged.[5–17]

In Turkey, monitoring and treatment of individuals with diabetes are carried out in polyclinics, which is a place where outpatients are treated. Most of the time, individuals with diabetes do not regularly attend health checks and are not monitored by health personnel. This lack of regular checkups stems from the fact that the polyclinic environment is crowded, and individuals with diabetes cannot reach the diabetes team whenever they need to. With the establishment of a Web site that will maintain diabetes training and monitoring on the Internet, individuals with diabetes could communicate with a healthcare provider whenever they want to and get the necessary support needed for diabetes self-care behavior.

As in all parts of the world, in Turkey, Internet use is becoming more common. In the Household Information Study conducted in the Seven Regions of Turkey, completed in April 2008, it was found that 24.5% of the families surveyed used the Internet.[18] This rate implies that the number of people using the Internet cannot be

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This study took an award of first oral presentation and good research at International Internal Medicine Congress in Antalya, Turkey (September 30 to October 4, 2009).

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underestimated, and Internet technologies could be used as effective tools for diabetes management.

In Turkey, the related institutes' Turkish Web sites intended for use by individuals with diabetes present the individuals with needed information, and some of them answer questions from those individuals. However, there is no information on how effectively diabetes patients use these Web sites or how useful the presented information is. Therefore, the aim of the study was to test the impact of Web-based diabetes training on the A_{1c} level and rate of health check attendance among individuals with type 2 diabetes.

RESEARCH HYPOTHESES

Two hypotheses were evaluated in this study: first, the A_{1c} levels are lower among individuals who receive diabetes training via the Internet compared with patients who receive diabetes training in a polyclinic setting; second, the health examination attendance rates of those who are trained on the Internet are higher than those of people who receive diabetes training in a polyclinic setting.

METHODS

Participants

The study population was composed of 122 individuals with type 2 diabetes, who could use the Internet, had completed their basic diabetes education, and shared similar basic situational factors. Using a randomization method, study participants were chosen using simple random sampling from patients who were being monitored by diabetes nurses. The sampling criteria for the experimental (n = 61) and control group (n = 61) were subjects had to be older than 18 years, had to have a diagnosis of type 2 diabetes at least 6 months earlier and undergoing insulin treatment, had to have completed basic diabetes education, had an A_{1c} level greater than 7%, without advanced level retinopathy and neuropathy, had to be able to use a computer and the Internet, and volunteered to participate in this study. By examining the medical reports of diabetic individuals, the exclusion criterion was to have advanced retinopathy and neuropathy; patients with either were excluded from the study.

Design

The research was designed to be experimental. The study was conducted at Dokuz Eylül University, in the Endocrine Polyclinic where there are four certified diabetes nurses who provide basic diabetes education; patients with diabetes are informed that they can come to the polyclinic whenever they need, participate in group training programs that are organized weekly, and should attend health checks every 3 months. There are 850 registered adults with type 2 diabetes who are expected to attend health checks every 3 months. Additionally, only 600 of these patients receive insulin treatment and have an A_{1c} level higher than 7%. When telephone calls were conducted to these 600 individuals with diabetes, 42 patients could not be reached, and 31 people indicated that they did not know how to use a computer or the Internet, and eight had advanced retinopathy and neuropathy. Therefore, 519 people with type 2 diabetes were found to be appropriate for the sampling features (Figure 1).

All of the individuals who were informed of the study and displayed appropriate sampling features chose to participate, and there was no attrition from the study. In total, 122 people were included in the study, and the sampling power was determined to be 85%. Power analysis was performed using two groups and the Minitab 14 program (Minitab Inc, State College, PA), Power and Sample Size menus. For the 122 selected patients, randomization was performed using the Minitab 14 package program in the computer environment. In the randomization, an experimental group and a control group were determined using simple random sampling. To determine homogeneity, the experimental and control groups were compared in terms of basic situational factors (ie, age, sex, training, marital status, A_{1c} level), and t and χ^2 tests were performed. The analyses demonstrated that there were no significant differences between two groups in terms of situational factors (Table 1).

After determining that the groups and the numbers did not differ significantly, the experimental group was trained on how to use the Internet program (September 15, 2008). Points that participants found to be unclear initially regarding the utilization of Internet were revised. Those who were struggling to use the program were retrained. Once the Internet program was opened for participant use, individual program usage was not monitored.

Measurements

To determine the effect of Web-based diabetes training on patient outcomes, the A_{1c} level and rate of attending health checks for each study participant were evaluated. The A_{1c} level results were obtained in a hospital setting using laboratory instruments tested by the hospital quality control team, who were accompanied by observers. The prestudy (November 2008) and poststudy (May 2009) data from the experimental group were gathered from the Web; the control group data were collected from the patient records at the endocrine polyclinic by a researcher.

Monitoring Site for Individuals With Diabetes

The computer engineering department aided in the construction of the Web site. The site-user profile was divided

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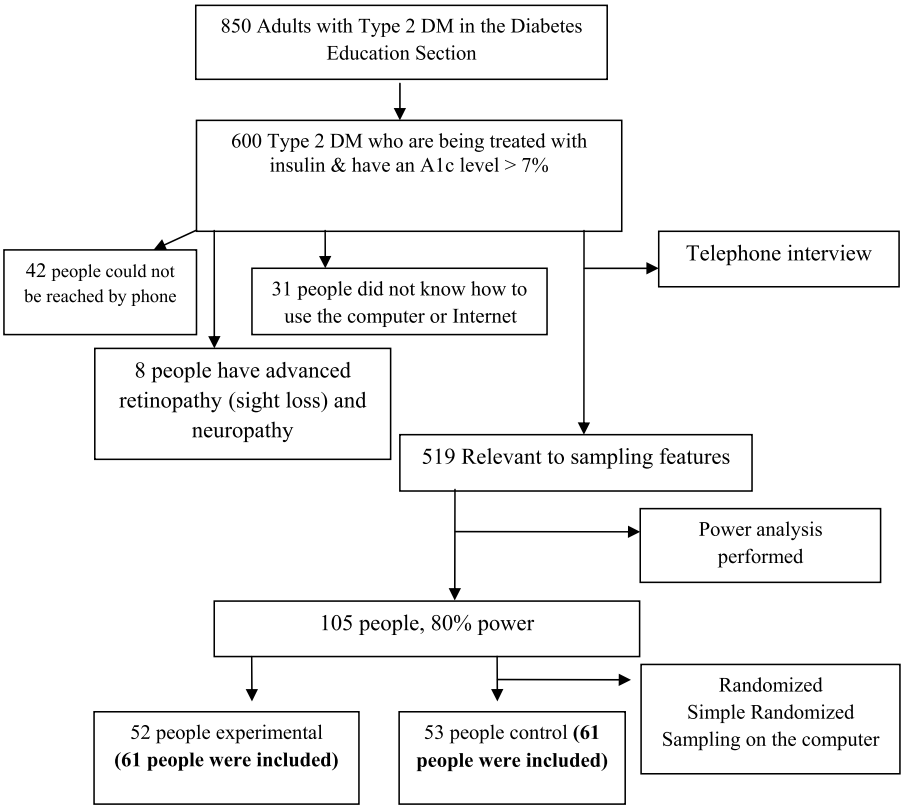

FIGURE 1. The stages of sampling and randomization.

into two categories, admin (diabetes nurse) and user (individual with diabetes). Microsoft Visual Studio (Microsoft, Redmond, WA) developed the software; C#.net 2005 was the programming language used, and a Microsoft SQL SERVER 2005 (Microsoft, Redmond, WA) was used as the database. To ensure that the interface was convenient to use, Microsoft Silverlight (Microsoft, Redmond, WA) and Ajax technologies were used. The site supports Microsoft Internet Explorer 6 and higher, as well as Mozilla Firefox 2 and higher.

Interventions

To define the individual characteristics of the individuals with diabetes, that is, age, sex, marital status, diabetes knowledge, glycemic control, blood fats, and body mass index, an identification form was prepared and uploaded to the Web site. After the patient filled in the questionnaire, he/she recorded the information on the record menu and sent it to the researcher through the "send it" menu. The researcher compiled all of the information specific to the user in a media file and sent the data back to user, who could view the information using a media player. The patients were able to read the files, download them to their computer, print and collect the files in a folder, and could ask the researcher questions through the message menu. The researcher answered the questions and then added them to the frequently asked questions on the Web so that all of the patients could read them.

In the admin menu, daily blood sugar monitoring was maintained by the researcher; blood sugar graphics specific to each patient were generated, which could also be viewed by the patient. While the experimental group was monitored after the previously mentioned Web-based education, the education and observations of the control group were provided by the diabetes nurse in the polyclinic environment. The education contents for the two groups were formatted according to American Diabetes Association 2009 criteria.

Statistical Analysis

The A_{1c} levels of the patients in the experimental and control groups were compared using an independent *t* test, and within-group comparisons at 0 and 6 months were done using a dependent *t* test. The comparison of those who attended checks and who did not was assessed using ratio test.[19] All the analyses were carried out with the Minitab 15 statistical package.

Research Ethics

This study was approved by the Dokuz Eylül University Nursing School Ethics Committee. Written and signed

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Table 1

Individual Characteristics of Diabetes Patients in the Experimental and Control Groups After Randomization (n = 122)

	Experimental Group (n = 61)		Control Group (n = 61)		
Individual Characteristics	**n**	**%**	**n**	**%**	**Homogeneity of Variances**
Sex					1.0000, $P > .05$
Female	31	50.82	31	50.82	
Male	30	49.18	30	49.18	
Marital status					0.4750, $P > .05$
Married	35	57.37	31	50.82	
Unmarried	26	42.62	30	49.18	
Educational level					0.2857, $P > .05$
Literate	3	4.91	5	8.19	
Primary school	12	19.67	9	14.75	
Secondary school	11	18.03	16	26.22	
High school	13	21.31	18	29.50	
Higher education	22	36.06	13	21.31	
Attending health checks	45	73.77	43	70.49	1.000, $P > .05$

	Experimental Group (n = 61)		Control Group (n = 61)		
	n	**Mean (SD)**	**n**	**Mean (SD)**	**Homogeneity of Variances**
Age	61	50.59 (7.5)	61	52.45 (7.01)	0.535, $P > .05$
A_{1c}	61	8.008 (0.97)	61	8.141 (0.98)	0.456, $P > .05$

informed consent was obtained from each participant. The principles developed by the Health on the Net Foundation (HON) for worldwide accreditation were used in the construction of the Web site.[20]

RESULTS

Data analyses demonstrated that there were no significant differences between the experimental and control groups in terms of regular health check attendance ($z = 0.00$; $P = 1.000$) at the start of this study. Six months after the study began, the rate of health check attendance was higher in the experimental group at 95% significance level ($z = 5.97$; $P < .05$), while there was no change in the control group ($z = -0.80$; $P = .426$) (Table 2).

Although subjects were randomly assigned to the experimental or control group, the pretraining A_{1c} level means did not differ significantly between the two populations at the 95% significance level ($t = -0.75$; $P = .456$). However, 6 months later, a significant decrease was recorded in the experimental group ($t = -3.97$; $P = .000$), while no difference was detected in the control group ($t = -0.63$; $P = .534$) (Table 3).

DISCUSSION

In this study, we found that patients monitored via the Internet had better heath check attendance rates than those monitored in the polyclinic environment. Diabetes can result in acute or chronic complications if glycemic control is not maintained. Therefore, patients need to have regular health checks to achieve their glycemic target and avoid possible complications. The American Diabetes Association[3] recommends regular monitoring of a patient's A_{1c} level, blood pressure, lipid level, and kidney function and sight tests, as well as regular neurological examination. If the A_{1c} level is lower than 7%, it is recommended that patients have their A_{1c} levels measured twice a year; diabetic individuals with A_{1c} levels of greater than 7% should have their A_{1c} levels measured once every 3 months.[3] In the current study, the health check attendance rate, which was 73.77% before the study, rose to 98.36% after the study; there was no change in the control group (Table 2). In the review of the literature, we found no

Table 2

Comparison of the Rates of Attending Health Checks Between the Two Groups (n = 122)

	Preenterprise	6 mo	
	n (%)	**n (%)**	***z* (*P*)**
Experimental group	45 (73.77)	60 (98.36)	5.97 (.000)
Control group	43 (70.49)	50 (81.96)	−0.80 (.426)
z	0.00	4.98	
P	1.000	.000	

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Table 3

Comparison of A_{1c} Levels Between the Groups (n = 122)

	Preenterprise	6 mo	
	Mean ± SS	Mean ± SS	*t* (*P*)
Experimental group	8.008 ± 0.972	7.496 ± 0.725	6.63 (.000)
Control group	8.141 ± 0.987	8.189 ± 1.153	−0.63 (.534)
t (*P*)	−0.75 (.456)	−3.97 (.000)	

published study that investigated the effects of Web use on patient health check attendance.

The mean A_{1c} level for the experimental group decreased to 7.496% from 8.008%; no change was seen in the control group mean of the A_{1c} level (Table 3). The review of literature identified eight studies that examined the impact of Web-based monitoring on a patient's A_{1c} level. These studies also found that the A_{1c} level of individuals with type 2 diabetes was lower in the Web-monitored group compared with the group monitored in the polyclinic.[12,21–30] In these reports, the effectiveness of the Internet-based training was explained by two factors, the healthcare system and the limited time of patients.[21,25]

In the current study, the effectiveness of Web-based training can be explained by the same reasons. In the Turkish healthcare system, each diabetes patient must make an appointment to reach a doctor or nurse; he/she must come to the institute to make an appointment and come back again for the examination. Also, because of time constraints, the patient may not express himself/herself adequately, or the health official may not sufficiently comprehend the patient's problem; sufficient time to start the treatment and the monitoring process may not be ensured. One of the difficulties in reaching health officials is the individual limits of the patient. Some are not able to leave their workplace because of their job responsibilities, and others cannot go out because of physical limitations. For this reason, Web-based monitoring use may be considered as a complementary tool for the monitoring process.

CONCLUSION

In this study, Web-based diabetes training positively impacted the A_{1c} level and heath check attendance. This finding demonstrates that Web-based training affects not only care outcomes, but also the monitoring process. Therefore, the results we obtained support the development of Web-based training sites for patient-centered enterprises, too.

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